

sortscore: Sort-seq MAVE scoring and visualization using Python

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Complete List of Authors:	Chitwood, Caitlyn ; Washington University in St Louis School of Medicine, Department of Pediatrics Orr, Emily ; Washington University in St Louis School of Medicine, Department of Pediatrics Baldrige, Dustin; Washington University in St Louis School of Medicine, Department of Pediatrics
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We provide our responses to the reviewers in-line below.

Reviewer: 1

Comments to the Author

Summary

In this work, Chitwood et al. present sortscore, a Python package for scoring sequence variants in Sort-seq-style, FACS-based DMS experiments. Specifically, sortscore implements a conventional scoring approach in which per-variant fluorescence levels are estimated as weighted means of binned read counts. The resulting scores can then be aggregated at the codon or amino acid level, with optional replicate and batch normalization. The package also includes built-in heatmap visualization.

The manuscript is clearly written, but I do have concerns about the novelty claim that no such tools exist for Sort-seq scoring. Several published tools already handle this task. That said, the statistical approaches that sortscore implements are reasonable given that they are straightforward and widely used in the field, and most groups still write their own ad hoc scripts to do such analyses. However, the authors need to justify the contribution relative to existing methods.

Major points

1. While sortscore provides a standardized implementation of a commonly used scoring scheme for FACS-based DMS data, tools using more sophisticated approaches for this type of analysis already exist. For example, Lilace [1] is a Bayesian hierarchical model developed for the same purpose and handles uncertainty explicitly. In addition, latent phenotype models like MAVE-NN [2] can also infer variant scores from the same type of data while accounting for both uncertainty and potential non-linearities in the measurement process. The authors do not discuss these tools or provide any comparison, making it unclear to me what additional value sortscore provides.

We thank the reviewer for their thoughtful comments on the sortscore manuscript, and we provide here our responses to the reviewer's concerns. Addressing these concerns has greatly improved this manuscript. We especially appreciate the reviewer bringing additional Sort-seq scoring tools to our attention. Our original novelty claim was based in part on a recent review of MAVE software (Çubuk et al., 2025), which did not identify any existing tools purpose-built for Sort-seq scoring. We recognize, however, that

Lilace, published shortly before our submission, begins to fill this gap, and we have revised the manuscript to remove the novelty claim and discuss Lilace's contribution (pages 4 and 15).

In this current revision, we also more thoroughly considered MAVE-NN, an additional tool suggested by the reviewer, as well as related tools MoCHI and LANTERN. The revised introduction now states that these tools fit global genotype-phenotype maps and report model-predicted latent phenotype values at the variant level (page 4). These tools provide valuable capabilities, but our assessment is that their primary purpose is not to calculate easily interpretable and independent variant scores. Our assessment is that sortscore and Lilace are more directly comparable to one another, as they both focus on estimating variant activity directly from each variant's binned read counts (via weighted mean and Bayesian hierarchical methods, respectively). These approaches provide variant activity estimates traceable to each variant's own originating data, which isn't directly available when using a globally fit model, like MAVE-NN and related tools. Our position is that this traceability remains important in contexts such as clinical interpretation of functional data.

We have now incorporated a new table (page 16) that strengthens the manuscript by comparing these methods. This takes the place of the previous software comparison table containing growth-based assay scoring methods, which has been moved to the Supplement.

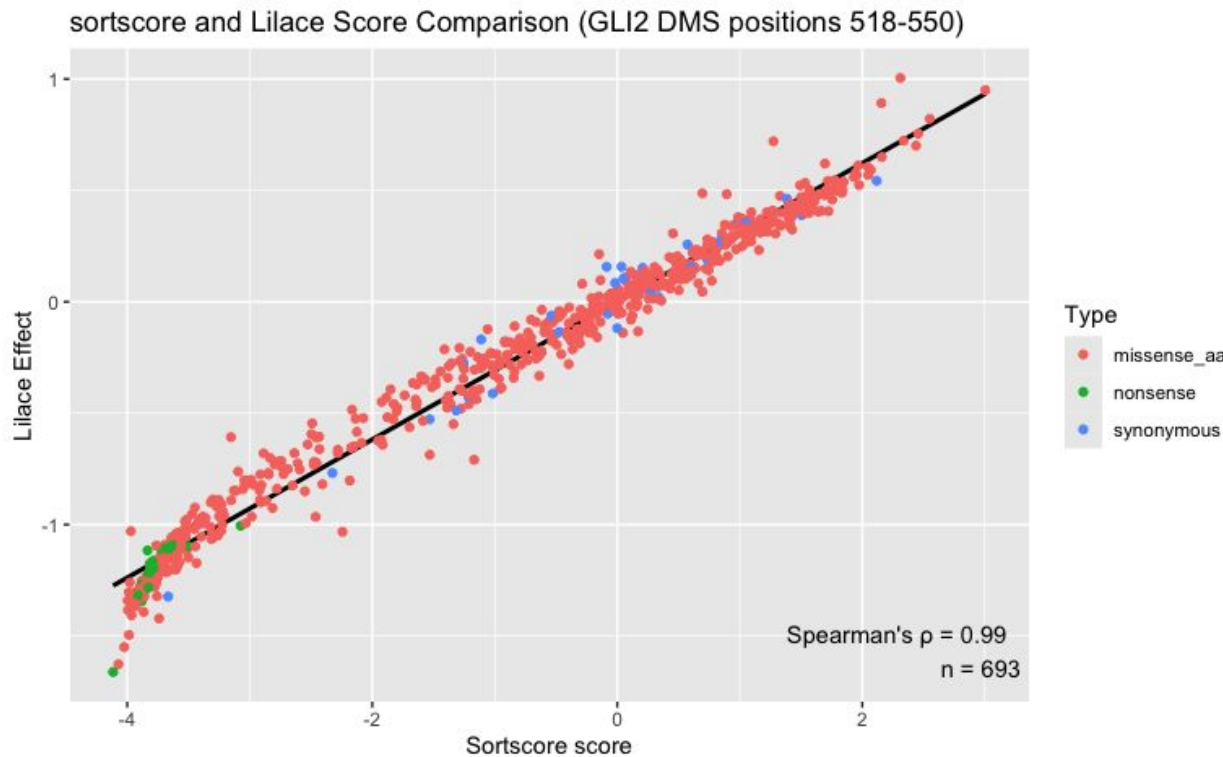
We have also now added a new demonstration notebook (https://github.com/dbaldrige-lab/sortscore/blob/main/demo_data/external_tools/Lilace/Lilace.md) in which Lilace was used to score a portion of the GLI2 DNA Binding Domain. This analysis approach uses a newly implemented integration command to export counts in a compatible format for use as input to Lilace, alongside sortscore output scores and metadata.

In addition, we have added a new figure to the Supplement (Supplementary Figure 6, also pasted below), which shows the expected high correlation between sortscore's weighted-mean approach compared to Lilace's Bayesian hierarchical scores (Spearman's $\rho = 0.99$, $n = 694$).

We note that Lilace plots do not accommodate results from codon mutagenesis currently. However, it's feasible that codon mutagenesis counts could be exported from sortscore, used to fit a Lilace model, and then plotted using heatmap functions from the sortscore API.

In the MAVE Background section on pages 3-5, we now address the utility of sortscore considering the wider context of Sort-seq tooling. Principally, that a reproducible implementation of weighted-mean scoring is valuable for reproducing historic analyses and reducing the burden of writing and reviewing custom scripts, given that the uptake

of Bayesian and neural net method uptake is ongoing. We also assert that sortscore is useful as a first pass scoring method in non-interactive HPC workflows, since it can be implemented through a bash one-line command. Finally, sortscore provides visualization options (i.e. 64 codon heatmap and tiled heatmaps) and normalization options across tiles that are especially useful in the context of short-read sequencing; these features do not appear to be implemented in other MAVE Sort-seq tools presently.



Supplementary Figure 6: Comparison of sortscore and Lilace variant-effect estimates. Variant scores from a FACS-based deep mutational scan of GLI2 residues 518–550 were analyzed using sortscore and Lilace. Each point represents one variant and is colored by mutation type. Estimates from the two methods were strongly correlated (Spearman's $\rho = 0.99$; n = 693 variants).

2. The authors argue that complex estimators “have the potential to inadvertently increase bias” in low-count settings, but this claim is not supported by any evidence. The Lilace paper directly benchmarks its method against the weighted-mean-bin approach and shows that the latter has a substantially higher FDR among low-count variants. I suggest removing or softening this unsupported critique.

We agree with this assessment upon review of Lilace benchmarking results and have removed this claim (page 9).

Minor points

1. sortscore requires MFI values for each sorted bin, but these are often unavailable in public datasets. I suggest either adding a fallback scoring mode using ordinal bin labels (e.g., 1, 2, 3, 4) or clarifying in the documentation that such labels can be used as a proxy when MFI values are unavailable.

This comment is consistent with Comment #4 from Reviewer 2, below. We have now clarified that sortscore is flexible to ordinal labels or other weights being input in the MFI column (page 4). We observed high correlations when using MFI or ordinal bin labels (see Supplementary Figure 3, also pasted in response to Reviewer 2, below). We also updated the usage guide (<https://github.com/dbaldrige-lab/sortscore/blob/main/docs/usage.md>) to specify that ordinal values can be input in cases where experimentally derived bin weights are not available.

Future versions of sortscore may capture these weights from multiple sources from which they can arise (e.g. FlowJo statistics and equipment printouts).

2. It seems like the section heading jumps to 4.2, skipping 4.1.

This has been corrected in the text.

References

- [1] Freudenberg, J., Rao, J., Howard, M. K., Macdonald, C., Greenwald, N. F., Coyote-Maestas, W., & Pimentel, H. (2026). Accurate variant effect estimation in FACS-based deep mutational scanning data with Lilace. *Genome biology*, 27(1), 48. <https://doi.org/10.1186/s13059-026-03934-1>
- [2] Tareen, A., Kooshkbaghi, M., Posfai, A., Ireland, W. T., McCandlish, D. M., & Kinney, J. B. (2022). MAVE-NN: learning genotype-phenotype maps from multiplex assays of variant effect. *Genome biology*, 23(1), 98. <https://doi.org/10.1186/s13059-022-02661-7>
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Reviewer: 2

Comments to the Author

Here Chitwood et al present sortscore, an integrated python package to analyze and visualize data from MAVE (multiplex assay of variant effect) experiments. This type of package is certainly needed, as MAVE data processing pipelines have tended to be bespoke to each study and have not been readily transferable between labs. Although other packages providing similar functionality have recently been introduced, sortscore addresses a specific type of MAVE which is not well addressed by these, namely MAVES using FACS (flow sorting) to bin variants by activity.

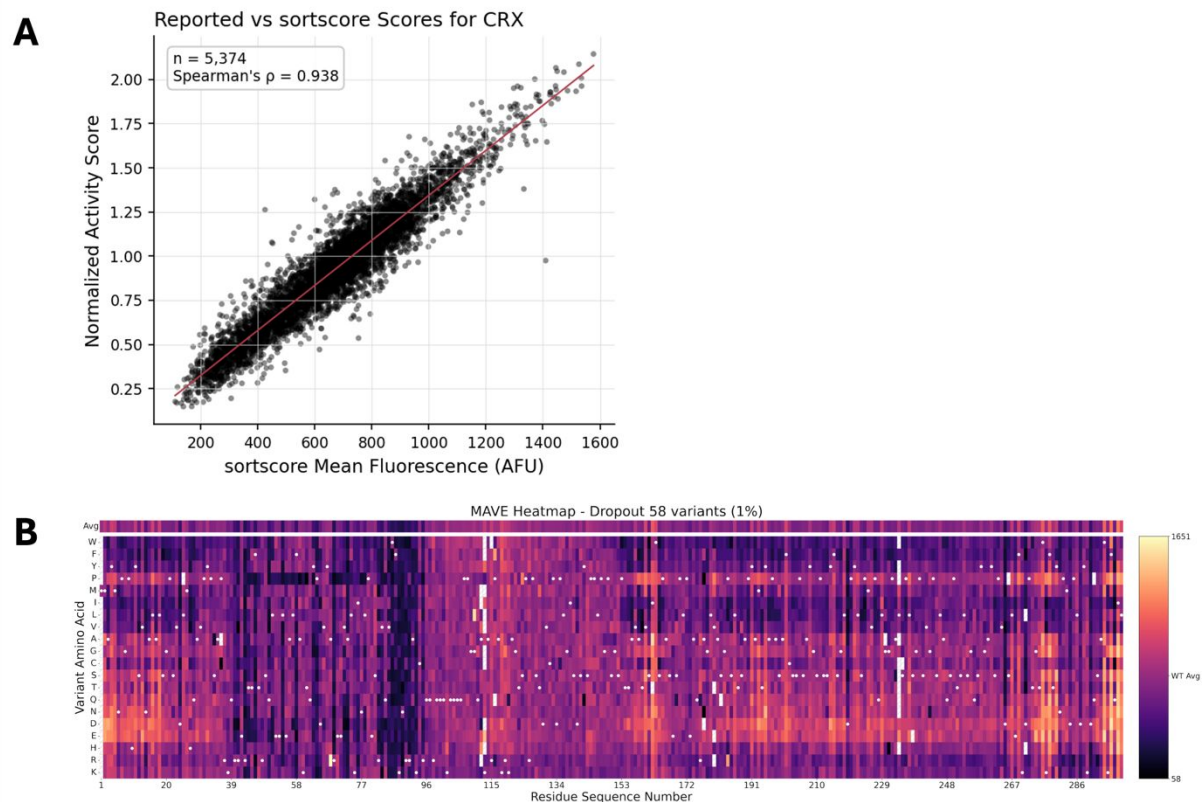
The proposed scoring model is straightforward and well justified, and the package is well documented and easy to use (I verified successful operation on the provided demo data). Sortscore tool will be a useful addition to the field, and this manuscript is clearly written and places the tool in the appropriate context. I have only a few relatively minor suggestions:

(1) In addition to the authors' GLI2 MAVE data, it would be important to show reanalysis of a previously published dataset. I would suggest either the CRX example cited, or published sort-seq based maps of HNF1A (DeForest et al, 2023; PMID 37492105) or PPARG (Majithia et al, 2016; PMID 27749844). Do the resulting scores differ substantially from what was reported in those maps?

Thank you for the positive feedback on our package! We especially appreciate that you were able to verify operation using the demo data! We also appreciate these suggestions for improvement, which have improved the manuscript significantly.

To demonstrate reanalysis of a previously published data set as requested, we have now used sortscore to analyze an additional DMS dataset, specifically the CRX example that we previously cited (Shepherdson et al.) (Supplemental Figure 5, also pasted below).

The results of the sortscore analysis and the published analysis are very similar to one another (Spearman's $\rho = 0.938$, $n = 5,374$). Default sortscore config options were used.



Supplementary Figure 5: Comparison of sortscore-derived scores (using default sortscore options) to reported scores for the CRX transcription factor DMS dataset (Shepherdson et al., 2024). **A:** Scatterplot of normalized reported activity scores versus sortscore mean fluorescence scores ($n = 5,374$; Spearman's $\rho = 0.938$). **B:** Heatmap showing sortscore-derived results for the CRX transcription factor.

(2) A main limitation I can see for sortscore is in the lack of statistical framework for estimating significance of each variant vs background. For instance, a z score or p value can help to identify if a given variant score represents a strong deviation from WT, or may instead be the product of low coverage, noise in a single replicate, etc. Some discussion of how this might be accommodated in the future would be welcome here.

Thank you for the opportunity to address these topics of distinguishing variant scores from background and incorporating measurement uncertainty.

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The current focus of sortscore is to reproducibly estimate variant scores rather than to formally quantify measurement uncertainty or test for significance. For these tasks and related tasks, such as determining an overall assessment of the assay’s ability to accurately discriminate benign from pathogenic variants, we have chosen to defer to analysis approaches developed by other groups, at this time. Some examples include the work by Brnich et al. and Zeiberg et al., which we have now cited in the manuscript on page 17.

As a related but distinct consideration, we provide quality control filters, including minimum counts (in reads per million), that allow users to remove low-confidence variants prior to performing downstream hypothesis testing or applying classification algorithms of their choice. These filters address data quality at the input stage, but we recognize the need for a statistical framework that distinguishes variant effects from noise and the background distribution. In future work we plan to provide measurement uncertainty metrics, including 95% confidence intervals and standard errors across replicates and codons when mutagenesis strategy allows. We note that these measures quantify the precision of an individual variant estimate but do not by themselves constitute a formal test of significance against WT or background. However, sortscore does calculate a modified z-score during batch normalization, and this could be adapted to provide a per-variant significance test, providing a p-value distinguishing significant difference from background as suggested by the reviewer. We have clarified that this is planned future work within the manuscript (page 17).

(3) Pg 9 – what is the standard approach for collecting median AFU from FACS bins? Often this is displayed in PDFs from FACS instruments (then, I guess, manually transcribed into the settings file), but it would be a very useful feature to extract these from FACS run metadata, FCS files, etc.

We agree that bin fluorescence values are an important reproducibility input. In the current version, these values are supplied explicitly in the experiment configuration file. We have clarified the expected source of this metadata in the documentation (<https://github.com/dbaldrige-lab/sortscore/blob/main/docs/usage.md>).

For future releases, import utilities are being considered that may capture these values from FlowJo statistics exports, pre-gated FCS files representing sorted bins, or instrument reports.

(4) Pg 9 – related to AFU units. How different are end results when using the actual AFU, versus synthetic values corresponding to bin rank (i.e., 1,2,3,4)

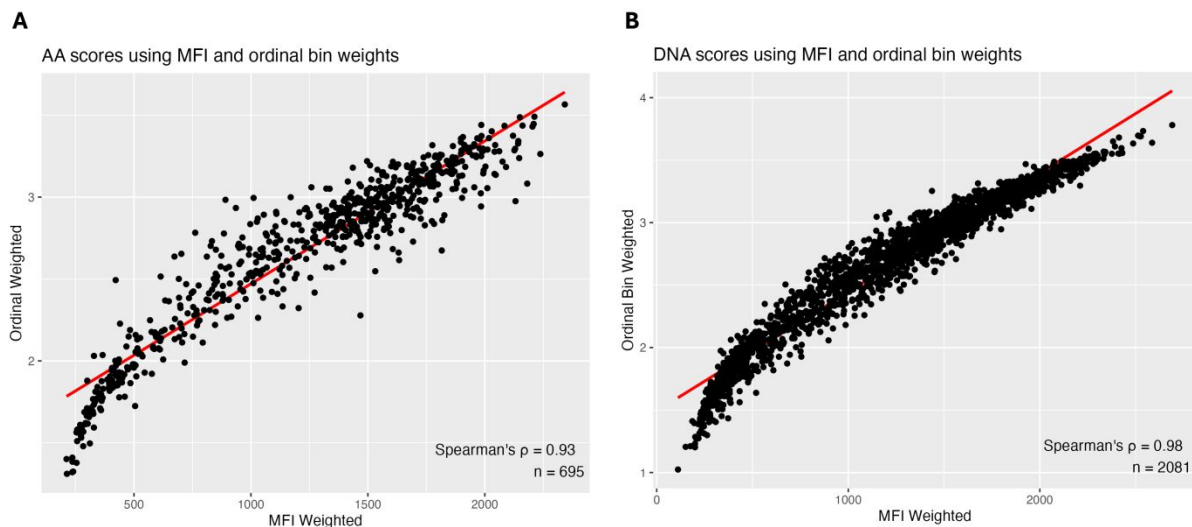
sortscore accepts any user-defined value in the MFI/weight column, so ordinal values can be substituted for actual fluorescence measurements when these are unavailable. In preparing this comparison, we debugged this option to ensure bin weights of 1,2,3,4 performed as intended.

We tested this option using one of the demo datasets previously provided (a FACS-based deep mutational scan of GLI2 residues 551–583), using both weighting schemes and comparing the resulting variant scores. Notably, median bin MFI values in this dataset were unevenly spaced (e.g. 125, 242, 953, and 3077 for replicate 1), yet concordance between MFI and ordinal-weighted scores remained high at both the amino-acid (Spearman's $\rho = 0.93$, $n = 695$) and DNA ($\rho = 0.98$, $n = 2081$) variant levels (Supplementary Figure 3). This suggests that ordinal weighting can serve as a reasonable substitute when MFI is not available, though we note that MFI-based weighting remains preferable when precise fluorescence measurements are available, since it captures actual differences in bin spacing that ordinal ranks do not.

The demo notebook used to generate this comparison is available at:

<https://github.com/dbaldrige->

[lab/sortscore/blob/main/demo_data/GLI2_oPool5b/ordinal_vs_mfi_weighted_scores.md](https://github.com/dbaldrige-lab/sortscore/blob/main/demo_data/GLI2_oPool5b/ordinal_vs_mfi_weighted_scores.md)



Supplementary Figure 3: Comparison of MFI and ordinal bin-weighted sortscore estimates. Variant scores from a FACS-based deep mutational scan of GLI2 residues 551–583 were calculated using either median fluorescence intensity (MFI) or ordinal

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values as bin weights. Each point represents an amino-acid variant (A: Spearman’s ρ = 0.93; n = 695) or DNA variant (B: ρ = 0.98; n = 2081).

(5) Pg 3 – all standard MAVE software has been designed to accommodate growth assay. I think this is generally true but it’s important to recognize exceptions, eg lilace and more general models such as MAVENN

We agree with this excellent point from the reviewer, which is similar to Major Point #1 from Reviewer #1, above. We now discuss the use of sortscore in the context of this wider tooling ecosystem that we were happy to have brought to our attention by both reviewers. A new demo notebook provides application of sortscore and Lilace to a portion of our lab’s GLI2 data to enable comparison of a selection of the standard outputs of each.

(6) Minor - Pg 16 line 23 “Atlases of Variant Effects” – don’t capitalize

The text has now been corrected.

sortscore: Sort-seq MAVE scoring and visualization using Python

Caitlyn Chitwood^a, Emily Orr^a, Dustin Baldrige^a

^aDepartment of Pediatrics, Washington University School of Medicine, St. Louis, MO, 63110, USA

Corresponding author:

Dustin Baldrige
Washington University School of Medicine
660 South Euclid Ave.
St. Louis, MO 63110
dbaldri@wustl.edu

ABSTRACT

Motivation

A growing number of tools enable the analysis of large variant libraries produced by multiplexed assays of variant effects (MAVEs). Experiments using fluorescent reporters and fluorescence-activated cell sorting sequencing (FACS-seq or Sort-seq) can coarsely quantify a given variant's impact on phenotypes such as transcription activity. Existing bioinformatics tools for Sort-seq data broadly fall into two categories: methods that model a genotype-phenotype landscape to infer latent variant phenotypes, and methods that directly estimate individual variant scores from experimental binned counts. Within this second category, sortscore provides an activity score computed directly from observed counts without fitting a model, retaining the original experimental scale when bin median values are known. ~~While several existing bioinformatics tools claim robustness for analyzing the binned data produced by Sort-seq, to date these have predominantly been applied to datasets produced by single measure, growth-based assays. To our knowledge, there are currently no publicly available tools that handle Sort-seq data natively.~~

Results

We present sortscore, a python package that incorporates a standard Sort-seq scoring method and normalization across sorted samples, technical replicates from separate sort times, and across oligos in tiled experiments. It also provides convenient heatmap visualizations. This software was used to analyze DMS experiments for the transcription factor GLI2. This work seeks to lower the barrier to entry and provide a clear starting point for scoring Sort-seq cell-based functional assays. This extends the availability of well-documented and reproducible data analysis protocols to a wider community employing MAVE techniques.

Availability

The package can be downloaded from PyPI using the pip installer. The code is also freely accessible and available for reuse through a Public GitHub repository (MIT License). Installation instructions, documentation, and tutorials are accessible at the sortscore GitHub repository: <https://github.com/dbaldrige-lab/sortscore>. [A snapshot of the code is available on Zenodo: https://doi.org/10.5281/zenodo.21724265](https://doi.org/10.5281/zenodo.21724265).

For Peer Review

1. Introduction

1.1 MAVE Background

Multiplexed assays of variant effects (MAVE) are cell-based assays used to measure experimental readouts of biological function and estimate the impact of thousands of genetic variants in a single experiment (Kinney, 2010). MAVE methods include deep mutational scanning (DMS), in which amino acid residues are systematically replaced, and via saturation genome editing, where single nucleotide variants are comprehensively evaluated (Tabet et al., 2022). The application of MAVE methods has grown rapidly since their introduction, in part due to their demonstrated ability to elucidate biological mechanisms through genetic perturbations (Esposito et al., 2019; Fowler & Fields, 2014) and attribute clinical outcomes to specific variants in a disease gene (Findlay et al., 2018; McEwen et al., 2026). Functional datasets also serve as promising gold-standards in the validation of computational variant effect predictors (Livesey & Marsh, 2020).

Growth phenotypes are the most common DMS experimental outcome, and most MAVE software developed to date has been designed to accommodate ~~some form of~~ growth assay (Çubuk et al., 2025). However, fluorescent reporters in combination with fluorescence-activated cell sorting (FACS) and subsequent DNA sequencing — referred to here as Sort-seq — are also employed in the MAVE literature (Aslanzadeh et al., 2025; Ma et al., 2024; Matreyek et al., 2018; Popp et al., 2025), including in massively parallel reporter assays (MPRAs) (Peterman & Levine, 2016). ~~Our Python package, sortscore, addresses the lack of dedicated Sort-seq DMS analysis software by explicitly processing the sequencing read counts data produced by these experiments and conveniently providing activity scores and visualizations.~~

1.2 Sort-seq Background

Sort-seq involves sorting a population of input cells into tubes defined by gating on a phenotype measured at single cell resolution, most commonly fluorescence

intensity level, followed by sequencing the DNA from those sorted cells. An overview of the Sort-seq workflow is shown in **Supplementary Figure 1**. This approach can be advantageous when a clinical phenotype or gene function may not be well correlated with cell proliferation and survival, and when fluorescent reporters are available to assess cellular function, such as reporters responsive to transcription factor activity (Shepherdson et al., 2024). Beyond functional testing genetic variants in the context of human health, Sort-seq methods have also been used to monitor bacterial populations (Lambrecht et al., 2017) and to estimate variant impacts on protein properties such as stability and solubility (Gaber et al., 2024). Sort-seq generally requires the incorporation of a fluorescent reporter into the biological pathway being studied, which may require cell line engineering.

The Sort-seq experimental process produces a count distribution for each population of cells carrying any one of potentially thousands of variants. ~~Each tube contains cells~~ Cells are sorted into discrete bins, each spanning a select ~~falling within a selected~~ interval out of the full dynamic range of the perturbed fluorescence phenotype (shown as bins in **Supplementary Figure 1.4**). Resulting sequencing files are processed to map sequencing reads back to mutagenesis variants. The sequencing read counts from each sorted sample reflect the relative abundance of each variant. These processed read counts become inputs for downstream analyses, such as those performed in our Python package, sortscore.

1.3 Sort-seq Scoring Software

Compared to growth assays, analytical software support for scoring Sort-seq MAVEs remains comparatively limited but is improving. The Lilace R package began filling this gap by providing an implementation of a Bayesian hierarchical scoring framework to estimate latent unimodal shifts in the fluorescence distribution for variants relative to a negative control baseline (e.g. synonymous distribution) (Freudenberg et al., 2026).

We developed sortscore, which implements a weighted mean scoring approach to provide an additional, dedicated, reproducible tool for Sort-seq analysis. It incorporates experimentally derived bin weights—in the case of FACS, median

fluorescence intensity (MFI)—when available, to retain quantitative detail and experimental units that are discarded when bins are treated as ordinal categories. Since MFI values are not strictly required, this approach can readily generalize to other bin-based MAVE assay designs beyond FACS-based Sort-seq. In its current implementation, sortscore processes sequencing read count data generated by FACS-based assays and provides activity scoring through a streamlined CLI workflow, along with visualization options (64 codon heatmaps and tiled heatmaps) and tiled normalization for short-read sequencing contexts—features which are not currently implemented in other Sort-seq MAVE tools.

While sortscore and Lilace directly score individual variants via weighted mean and Bayesian hierarchical methods respectively, a branch of related tools models global genotype-phenotype landscapes from which variant-level phenotypes are then inferred. MAVE-NN is one such tool which accepts binned counts to train a neural network that deconvolutes measurement noise and nonlinearities (Tareen et al., 2022). LANTERN and MoCHI apply low-dimensional and mechanistic biophysical approaches, respectively (Tonner et al., 2022; Faure and Lehner, 2024). The methodological differences between these tools are described further in **Table 1** below.

Beyond these methodological distinctions, the customizable workflows implemented by sortscore lower barriers to Sort-seq analysis. Since weighted mean scoring has been commonly applied historically, sortscore facilitates reproduction of these prior analyses within a standardized framework. Its CLI implementation further provides an accessible option for researchers working in high-performance computing environments or incorporating scoring into a larger data pipeline.

2. Methods

2.1 Software Design

Standardized Analysis Workflows

Figure 1 (below) shows two standardized analysis workflows that can be executed using the listed bash commands. These console commands offer a convenient way to add scoring and visualization to the end of a sequencing data processing pipeline, as in compute environments with batch job schedulers. The Variant Scoring workflow takes processed sequencing counts files from Sort-seq experiments and returns a heatmap and a Score Set CSV file. ~~This fills the gap in dedicated software and documentation for scoring Sort-seq experiments.~~ Prior to normalization, batch effects can be apparent across Sort-seq experiments. The Batch Normalization workflow takes Score Set files from multiple tiled experiments, and returns a combined Score Set and heatmap on a unified scale for comparison.

Dependencies

The sortscore package utilizes standard scientific Python libraries including NumPy and pandas for data processing, SciPy for statistical analysis, matplotlib and seaborn for visualization, and Biopython for biological sequence handling (Cock et al., 2009; Harris et al., 2020; Hunter, 2007; The pandas development team, 2026; Virtanen et al., 2020; Waskom, 2021).

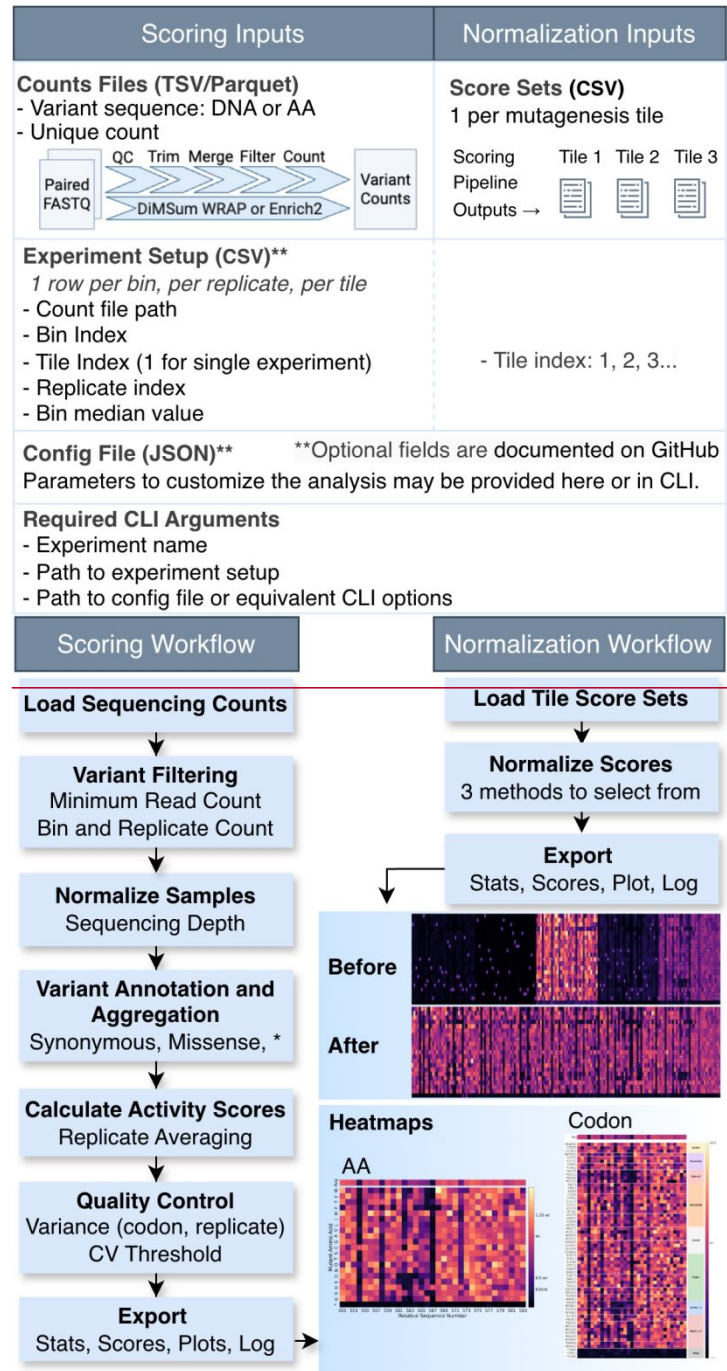
Data Loading and Input Requirements

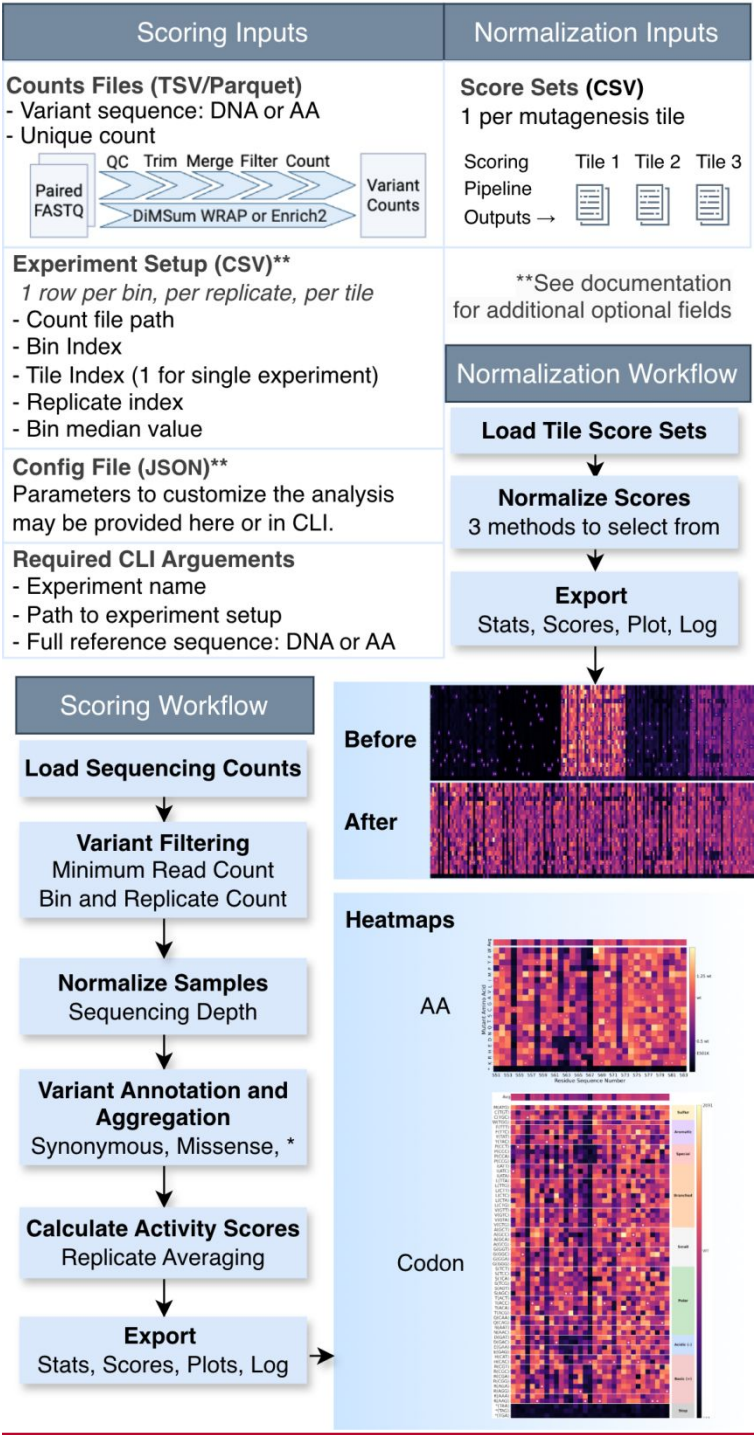
The sortscore package is focused on calculating variant scores based on relative abundance in sorted bins, normalizing data across tiled sorting experiments, and visualization of these results via heatmaps. Sortscore accepts variant counts files in TSV (gzipped or uncompressed) and Parquet formats, requiring that they contain two columns: sequence (or sequence identity) and unique counts. Currently available

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software is routinely used to convert raw sequencing files into the input variant counts file. Such tools include Enrich2 or DiMSum’s WRAP module for short reads, or Pacybara for long reads (Faure et al., 2020; Rubin et al., 2016; Weile et al., 2024). Software for mapping barcodes to variants is available in the Call Variants DMS Docker container (*P-Ruzycki/Call-Variants-DMS-Docker*).

Variants with low read counts ~~are may be~~ filtered during loading by specifying a threshold in units of reads per million in the configuration file ~~to remove counts that are less than three, though this threshold is adjustable in the analysis configuration files described below and in the package documentation.~~ This minimum read filter applies at the bin level and is 0 by default.





2.2 Analysis Configuration

Configuring Scoring for a Single Sorting Experiment

Easily modifiable configuration files provide flexibility to tailor the analysis workflows to new experimental assays using a basic text editor. Configuration for running sortscore is provided via two files: a) a JSON configuration file, and b) a CSV file to define the experimental design. Templates for each are available on GitHub: config.json and experimental_setup.csv, respectively. The project documentation on GitHub also lists the full set of required and optional fields available to customize the scoring workflow.

Modifiable fields in the JSON include the mutagenesis approach (e.g., all possible amino acids per position, presence or absence of nonsense variants), quality filter thresholds, and visualization customizations. The experiment setup CSV file provides the structure to share file paths, information about technical replicates, sum of read counts per sample, and gating strategy for a single sorting experiment.

Configuring Normalization Across Tiled Experiments

The same configuration file can be used to select the desired normalization method, specify negative controls to use during normalization, and provide paths to the configuration files for individual experiments that will be combined. The three available methods for normalization are described below. Prior to normalization across tiles, count files for each tile must be processed using the Variant Scoring workflow.

DNA and AA Variant Scoring and Heatmaps

Since a wide range of mutagenesis approaches are used in MAVE experimental designs, we provide a limited set of variant aggregation and heatmap customization options. When DNA sequences or variant identifiers are provided, the user can specify whether they want scores and heatmaps returned that reflect the DNA-level variant or protein sequences that have been aggregated across codons. The package auto-

detects the type of sequences present in the counts files and will alert the user if this is incompatible with the requested scoring option.

2.3 Variant Activity Scoring

The primary function of the first sortscore analysis pipeline is to return an activity score for each binned variant. To quantify activity, we estimate the mean phenotypic value utilizing elements from the discrete counts distribution for each variant. In Sort-seq, these distributions are produced by sorting cells into tubes corresponding to intervals of the measured range of phenotype values followed by sequencing. This simple estimator is practical, easy to understand and compute, and is useful for ranking variants and comparing resulting scores against the activity of benign-like and/or pathogenic-like control variants. ~~More complex estimators and modeling of parameters have their own demonstrated benefits, some of which are explored in softwares listed in Table 1. Due to the low counts often seen in MAVE experiments, along with the limited number of bins available in Sort-seq assays employing FACS, such tools have the potential to inadvertently increase bias and reduce interpretability by clinicians and curators looking to make use of the analysis results. This concern led us to opt for a direct calculation of mean fluorescence intensity, using binned counts and bin median values, as our output measure.~~

Normalization across Samples

The package controls for sequencing depth differences by normalizing total read counts across samples as reads per million (RPM). Total reads in a sample can be user-provided or can be calculated from the counts files.

$$RPM_{v,b} = \left(\frac{\text{read count}_{v,b}}{\text{total reads}_b} \right) \cdot 10^6$$

v = Variant index

b = Bin index

Score Calculation Estimating Variant Fluorescence

For each variant, Sort-seq produces a discrete distribution defined by ordered fluorescence bins and the counts of variant sequencing reads observed in each bin, which can be modeled as a probability mass function. FACS discretizes the underlying continuous fluorescence distribution of the cell population containing a given variant (see **Supplementary Figure 2**). The sortscore package calculates an expected fluorescence score, S_v , score for each variant under this discrete model as a weighted mean across bins, weighting the proportion of variant reads in each bin by a user-provided bin weight and summing across bins. This weight is typically the bin MFI for experiments employing FACS, but any user-defined value can be substituted, including an-by-weighting by an integer-based ordinal value.; MFI-based weights capture the actual, and often unevenly spaced, differences in intensity between bins, whereas ordinal weights assume evenly spaced ranks. Despite this simplification, we found ordinal-weighted scores to be highly concordant with MFI-weighted scores in a demo dataset with unevenly spaced bins (see **Supplementary Figure 34**). which serves as an estimate of MFI-based bin weights produce an estimate of the variant-carrying cell population's mean fluorescence. Ordinal bin weights produce a comparable rank-based score directly from binned counts, without fluorescence units. Both options serve as proxies for variant function. the mean fluorescence of the cell population and a proxy for variant function.

$$S_v = \sum p_{v,b} w_b \quad S_v = \sum_{b=1}^n M_{AFU,b} \cdot p_{count,b}$$

$M_{AFU,b}$ = Median fluorescence intensity of bin b in arbitrary fluorescence units (AFU)

~~$p_{count,b}$ = Proportion of variant reads (normalized for sample sequencing depth) in bin b~~

n = Number of sorted bins

w_b = median fluorescence intensity or ordinal value assigned to bin b ,

$p_{v,b}$ = proportion of reads for variant v in bin b after normalization for sequencing depth

Technical Replicate Averaging

By default, when averaging scores across technical replicates (defined as the same starting cell population sorted into bins in more than one FACS session), variant read counts serve as weights for calculating the mean across replicates. We use this weighting because we assume that higher read counts in a given replicate will result in more accurate scores for that replicate. The user can opt for a simple average instead in the analysis configuration.

$$\overline{S}_{v,\text{weighted}} = \frac{\sum_{r=1}^R S_{v,r} \cdot w_{v,r}}{\sum_{r=1}^R w_{v,r}}$$

- $S_{v,r}$ = Activity score for variant v in replicate r
- $w_{v,r}$ = Total normalized read count for variant v in replicate r
- R = Number of replicates

Options for Normalization and Scaling across Tiled Experimental Batches that Use Tile-Based Mutagenesis

Our package provides two options for reducing potential batch effects that may be observed across different tiled experiments, by which we mean the use of separate oligos that code for different portions of the same protein. These score adjustments to the activity scores bring them onto a similar scale for comparison.

Method 1 Linear range normalization

We provide this normalization method used in the 2018 publication of the BRCA1 saturation genome editing DMS (Findlay et al., 2018). An interval scale is set to the score difference between global nonsense (zero) and global synonymous score (one) to bring batches into alignment and aid interpretability.

$$S_{\text{norm}} = b \left(\frac{A - C}{a - c} \right)$$

b = Variant activity score

$\left(\frac{A - C}{a - c} \right)$ = Scaling factor (global range / local range)

a = Synonymous median (local)

c = Nonsense median (local)

A = Synonymous median (global)

C = Nonsense median (global)

Method 2: Biological two-control anchoring, “two-pole” (Default)

This normalization method (see **Supplementary Figure 43**) is the default used by sortscore and assumes that the tile-specific synonymous variant scores and nonsense variant scores should be similar to one another across experiments. It involves a three-step process: 1) normalization by the ratio of the global median (M_{syn}^{global}) and the tile-specific median (M_{syn}^{tile}) of synonymous variants, which all code for the same wildtype protein, 2) standardization of scores based on the scale of the distribution of synonymous scores, and 3) normalization by the ratio of the global and tile-specific medians of nonsense variants. The units of scores resulting from this method do not represent either the relative fold change in activity or the unit variance

from a particular distribution, but rather should be interpreted in reference to positive and negative controls, when possible. However, this method does cause the directionality of activity change to be immediately apparent, because the mean synonymous score is centered at zero.

$$S_v^{(1)} = S_v \cdot \frac{M_{syn}^{global}}{M_{syn}^{tile}}$$
$$S_v^{(2)} = \frac{S_v^{(1)} - \bar{S}_{syn}^{(1)}}{\sigma_{S_{syn}^{(1)}}}$$
$$S_{norm,v} = S_v^{(2)} \cdot \frac{M_{nonsense}^{global}}{M_{nonsense}^{tile}}$$

Method 3: Biological single control anchoring, “one-pole”

This method is equivalent to Method 24, except that it omits the nonsense normalization factor to accommodate experiments that may not include multiple nonsense variants in each experimental batch. Instead, only the synonymous scaling factor is applied to scores by multiplying by the ratio of the global synonymous median and the tile-specific synonymous median.

Quality Control Filtering Options

In an effort to improve the quality of the resulting scores, we provide options for filtering out or dropping values that may not meet certain thresholds. For example, a minimum replicates filter allows users to specify the number of technical replicates that each variant should be present in so that it can be included in the final score set of values averaged across replicates. The rationale for this filter is that if a variant is missing from multiple technical replicates, it may be poorly represented in the studied population, and any resulting scores may be unreliable due to insufficient sampling of that variant. A related minimum bins filter requires that a variant is represented in a given number of sorted samples in order to be assigned a replicate level score. This filter is highly context dependent because some experimental designs will intentionally

yield zero counts in some bins, but as a guide one might consider the number of bins that control variants fall into. The default behavior is to not apply these filters.

~~An additional coefficient of variation (CV) quality filter is provided to allow elimination of variants that have unacceptably high variation across repeated measurements. The CV measure is partitioned into variation across replicate and across codons in the case of DNA level mutagenesis that incorporate multiple codons per amino acid into their library. The default value is set to 0.5 and adjustable.~~

2.4 Data and Visualization Outputs

Plotting Capabilities

A potentially particularly valuable feature to users is that sortscore produces codon and protein substitution heatmaps to visualize relative activity scores across variants. Position within the protein is oriented on the y axis and each variant is represented by a different row on the x axis. Optional biophysical property annotations may aid in interpretation of calculated scores. Positional average scores and coverage statistics are also included in the heatmap visualizations. ~~Options include generating replicate scatterplots to visualize reproducibility and activity score histograms; these are included in the HTML summary report if that output option is selected. In the default data analysis pipeline, only activity score heatmaps are plotted automatically to avoid cluttering the user's environment.~~ Plotting functions can save outputs in PNG, PDF, or SVG formats.

Data Exports

Sortscore also returns full Score Sets, Score Set summary statistics, and the matrix of scores that are displayed in the heatmap. The heatmap matrix values are provided so the user can easily recreate and reformat the plot using other tools. This output will be raw scores if sortscore was run separately for individual experiments, or adjusted -scores if run in batch mode. Summary statistics are also provided for different classes of variants in a JSON file, such as synonymous with wildtype versus nonsense

versus all missense variants, allowing rapid assessment for how the scores were distributed in a given experiment.

2.5 MAVE Ecosystem Integration

MaveDB serves as a centralized repository for functional datasets and provides visualization functionality, with upcoming ClinVar submission integration (Esposito et al., 2019). Structured JSON configuration files and detailed output and error logging in sortscore enhance its interoperability with web applications, enabling potential integration with community resources such as MaveDB.

3. Results

3.1 Processing Sort-seq Data from GLI2 DMS

The data shown in the heatmap in Figure 1 ~~(and provided in Supplementary Figures 1 and 2)~~ was produced in a series of DMS experiments performed by our lab with the aim of characterizing the pathogenicity of variants within the GLI2 transcription factor DNA binding domain, from a manuscript in preparation. Additional experimental details are provided in Supplementary Figures 1 and 2.

A fluorescent reporter integrated into the genome of the NIH3T3 cell line produced GFP as a readout of intact transcriptional activity (Pusapati et al., 2018). GFP (FITC) fluorescence intensity was used to divide the cell population into four equal bins, each representing approximately 25% of the GFP expression distribution (see Supplementary Figure 2). ~~Cells were quantile sorted into 4 binned samples (25% per bin) according to fluorescent intensity, and~~ DNA was then extracted and sequenced. and f-low cytometry data were analyzed using FlowJo software.

Counts data were produced from FASTQ files using standard bioinformatics tools: cutadapt for trimming adapters (Martin, 2011), VSEARCH for merging paired reads (Rognes et al., 2016), and a custom awk script used to confirm the read was a member of the list of designed sequences. As expected, the sortscore analysis prominently displays the deleterious impact of variants in the zinc finger domain as

vertical dark stripes on the heatmap, particularly for the residues known to chelate the zinc ions (Pavletich & Pabo, 1991).

The use of a sequential colormap for the heatmap improved interpretability of these GLI2 data compared to the standard red-blue versions commonly seen in MAVE publications which have great utility in the presence of both gain and loss of function missense variants. However, we have found no definitive evidence for gain of function behavior in the GLI2 DNA binding domain DMS to date. When only one prominent direction of effect is present, as in these data, sequential colormaps can make critical residues more apparent. The complete dataset will be made available upon publication of a manuscript currently in preparation, which also utilized sortscore for analysis.

We also tested the package by scoring an existing DMS dataset for the CRX gene using default options (Shepherdson et al., 2024). The resulting heatmap is shown in **Supplementary Figure 56**.

3.2 Tool Comparison

Table 1 considers publicly available software packages that address the need for variant scoring in Sort-seq experiments. These separate broadly into two approaches, 1) genotype-phenotype landscape modeling methods that first model a surface for the region then predict latent phenotypes as a function of this feature space and 2) methods directly utilizing individual variant fluorescence distributions from experimental binned counts to produce estimates. In **Table 1**, we summarize some tools presently available for MAVE analysis. None of these publicly available software packages currently address the need for Sort-seq scoring, despite a large variety of tools optimized for analyzing growth-based assays. The sortscore package adds to this ecosystem by fills this gap by providing providing an intuitive and straightforward-direct activity score calculations in a user-friendly and well-documented package. When the experimental bin median values are known, scores are returned in the original scale of the experiment that generated the data.- Of the tools considered, Lilace is the closest comparator to sortscore in its goal of providing a dedicated tool for scoring FACS-based DMS data; we therefore compared sortscore estimates to those from Lilace on a FACS-

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based deep mutational scan of GLI2 residues 518–550, finding strong agreement between the two methods (Spearman's $\rho = 0.99$, $n = 693$; **Supplementary Figure 65**). In **Supplementary Table 1**, we summarize additional tools, primarily designed for growth assays, that are presently available for MAVE analysis and may be of interest to the reader.

For Peer Review

Visit <https://www.varianteffect.org/analysis-tools/> to view a curated list of MAVE software and supported applications. Upon review, we did not find any of the listed tools capable of performing continuous scoring from binned samples. This limitation was confirmed by discussion with the maintainers of TileSeqMave, DiMSum, and Rosace, and in a recent review (Çubuk et al., 2025).

For Peer Review

Table 1: Comparison of a selection of selected MAVE analytical tools that provide variant-level activity scoring or phenotype prediction for Sort-seq experiments.

Software	Assay (Fitness Estimate)	Method	Uncertainty Quantification
Sortscore (This paper)	Sort-Seq Deep Mutational Scan (Mean fluorescence)	Counts-weighted average	Standard error decomposition by replicate and codon
FORECAST (Gilliot & Goroehowski, 2023)	Sort-Seq Massively Parallel Reporter Assay (Mean fluorescence)	Maximum Likelihood Estimation / Method of Moments	Confidence intervals from Fisher information, delta method
Enrich2 (Rubin et al., 2016)	Growth (Log enrichment ratio)	Mixed effects model	Random effects term
DiMSum (Faure et al., 2020)	Growth (Log enrichment ratio)	Replicate enrichment with interpretable error model	Error model with additive and multiplicative terms
Resace (Rao et al., 2023)	Growth (Log enrichment ratio)	Bayesian shrinkage	Posterior intervals
TileSeqMave (Gersing et al., 2024; Weile & Roth, 2018)	Growth (Log enrichment ratio)	Error-weighted averaging	Bayesian refinement or regularized standard error

Software	Type	Summary	Variant Score / Phenotype Prediction	Uncertainty Quantification
<u>sortscore</u>	<u>Variant Scoring from binned counts</u>	<u>Quickly score variants and align tiled Sort-seq experiments</u>	<u>MFI bin weights produces a variant fluorescence estimate; ordinal bin weights produce a rank-based score</u>	<u>Not presently implemented</u>
<u>Lilace</u>		<u>Robust FDR control for DMS employing FACS</u>	<u>Latent unimodal shift in fluorescence distribution</u>	

			<u>relative to negative control</u>	<u>local false sign rate (lfsr)</u>
<u>MAVE-NN</u>	<u>Genotype-Phenotype Landscape Modeling</u>	<u>Learn quantitative G-P maps and deconvolute noise and nonlinearities</u>	<u>Output of the nonlinear function transforming the latent phenotype into measurement units</u>	<u>Parametric bootstrap approach</u>
<u>LANTERN</u>		<u>Find low-dimensional feature space for interpretable discovery</u>	<u>Posterior Mean of the nonlinear surface</u>	<u>Posterior predictive uncertainty</u>
<u>MoCHI</u>		<u>Fit mechanistic models to quantify absolute energies and interactions</u>	<u>Predicted observed phenotype that converts molecular phenotype into units of assay</u>	<u>Monte Carlo confidence intervals</u>

4. Limitations and Future Development

4.1 Opportunities for Improvement

Sortscore is focused on addressing scoring and visualization needs for Sort-seq experiments ~~scoring~~ and avoids redundant implementation of functionality already provided by other, more established MAVE software. At present, While this requires users to independently must generate input countss files from raw sequencing filesdata prior to running sortscore, though future integration improvements work couldmay streamline streamline that workflowhandoff from tools that perform this upstream step. This initial release implements an easy-to-interpret weighted average and basic quality filtering. Because this initial implementation computes scores directly rather than fitting a statistical model, uncertainty quantification is not built into the estimation method; as a result, sortscore outputs are best used as inputs for downstream MAVE interpretation rather than as a final analysis endpoint at this time. Additional next steps may include identifying variants with activity distinguishable from a reference background distribution, comparing score distributions across functional classes, integrating external annotations, and prioritizing variants for follow-up testing. Toward this end, sortscore already calculates a modified z-score during batch normalization, and this internal machinery could be extended in future work to provide a formal per-variant significance test, such as a p-value distinguishing variant activity from background.

Examples of analysis approaches that are adjacent to the sortscore package include the Oddspath calculations which can both be used to determine an overall assessment of a functional assay's ability to accurately discriminate benign from pathogenic variants (Brnich et al, 2018), and ExCALIBR, which can assign evidence levels to individual functional scores (Tejura et al., 2026). At a higher level, community-maintainable and reproducible approaches for bundling that bundle MAVE analysis tooling might include packaging containerizing complementary MAVE software supporting end-to-end analysis software via a Docker container or developing a Nextflow modules that could be used alongside with the deepmutscan nf-core pipeline (Wehnert & Stammnitz, 2025/2025).

———— This initial release of sortscore currently implements an easy to interpret weighted average and basic quality filtering. Error propagation through the batch normalization is not included at present. This and more complex estimators of scores are potential additions for future updates.

4.2 Looking Beyond FACS: Applicability across Cell Sorting

Unexplored and underutilized sortable cellular phenotypes show potential for adding orthogonal layers of evidence to atlases of variant effects. Additional multiplexed measurement modalities compatible with sequencing would be valuable for directly quantifying variant effects and mapping their impact onto underlying biological mechanisms. Current sorting approaches coarsen phenotypic data, introducing significant noise at this stage of Sort-seq experiments. Throughput of millions of cells containing thousands of variants is possible in a single experiment, but even so, profiling variants using Sort-seq techniques remains costly, both in time and reagent / equipment expense. Yet the clear conceptual throughline from MAVE data generation to diagnostic yield primes these techniques to be substantially sharpened in the coming years.

Sortscore could be applied to score any continuous, sortable phenotype, so long as the technique doesn't degrade the binned genetic material. This package could

provide an easily accessible scoring framework for up and coming technologies that may expand what we refer to as Sort-seq. Sortable and multiplexable phenotypes might include additional spectroscopic, electrical, mechanical, or morphological cellular features. Responsibility lies with the user to determine if such a metric adequately captures the genotype-phenotype relationship.

4.3 Concluding Remarks

We provide a clearly documented and accessible tool to serve as a starting point for analyzing Sort-seq counts data. Questions, bug reports, and feature requests may be directed to the project's GitHub Issues page at <https://github.com/dbaldrige-lab/sortscore>. By releasing this package, we hope to lower the barrier to entry for groups exploring these experimental methods. We also hope to make validating the conclusions gleaned from Sort-seq equally approachable to groups that do not routinely use these methods, which may increase parity in the analysis protocols for Sort-seq variant effect assays compared to the more widely employed growth-based methods in use today.

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For Peer Review

sortscore: Sort-seq MAVE scoring and visualization using Python

Caitlyn Chitwood^a, Emily Orr^a, Dustin Baldrige^a

^aDepartment of Pediatrics, Washington University School of Medicine, St. Louis, MO, 63110, USA

Corresponding author:

Dustin Baldrige
Washington University School of Medicine
660 South Euclid Ave.
St. Louis, MO 63110
dbaldri@wustl.edu

ABSTRACT

Motivation

A growing number of tools enable the analysis of large variant libraries produced by multiplexed assays of variant effects (MAVEs). Experiments using fluorescent reporters and fluorescence-activated cell sorting sequencing (FACS-seq or Sort-seq) can coarsely quantify a given variant's impact on phenotypes such as transcription activity. Existing bioinformatics tools for Sort-seq data broadly fall into two categories: methods that model a genotype-phenotype landscape to infer latent variant phenotypes, and methods that directly estimate individual variant scores from experimental binned counts. Within this second category, sortscore provides an activity score computed directly from observed counts without fitting a model, retaining the original experimental scale when bin median values are known.

Results

We present sortscore, a python package that incorporates a standard Sort-seq scoring method and normalization across sorted samples, technical replicates from separate sort times, and across oligos in tiled experiments. It also provides convenient heatmap visualizations. This software was used to analyze DMS experiments for the transcription factor GLI2. This work seeks to lower the barrier to entry and provide a clear starting point for scoring Sort-seq cell-based functional assays. This extends the availability of well-documented and reproducible data analysis protocols to a wider community employing MAVE techniques.

Availability

The package can be downloaded from PyPI using the pip installer. The code is also freely accessible and available for reuse through a Public GitHub repository (MIT License). Installation instructions, documentation, and tutorials are accessible at the sortscore GitHub repository: <https://github.com/dbaldrige-lab/sortscore>. A snapshot of the code is available on Zenodo: <https://doi.org/10.5281/zenodo.21724265>.

1. Introduction

1.1 MAVE Background

Multiplexed assays of variant effects (MAVE) are cell-based assays used to measure experimental readouts of biological function and estimate the impact of thousands of genetic variants in a single experiment (Kinney, 2010). MAVE methods include deep mutational scanning (DMS), in which amino acid residues are systematically replaced via saturation genome editing, where single nucleotide variants are comprehensively evaluated (Tabet et al., 2022). The application of MAVE methods has grown rapidly since their introduction, in part due to their demonstrated ability to elucidate biological mechanisms through genetic perturbations (Esposito et al., 2019; Fowler & Fields, 2014) and attribute clinical outcomes to specific variants in a disease gene (Findlay et al., 2018; McEwen et al., 2026). Functional datasets also serve as promising gold-standards in the validation of computational variant effect predictors (Livesey & Marsh, 2020).

Growth phenotypes are the most common DMS experimental outcome, and most MAVE software developed to date has been designed to accommodate growth assays (Çubuk et al., 2025). However, fluorescent reporters in combination with fluorescence-activated cell sorting (FACS) and subsequent DNA sequencing — referred to here as Sort-seq — are also employed in the MAVE literature (Aslanzadeh et al., 2025; Ma et al., 2024; Matreyek et al., 2018; Popp et al., 2025), including in massively parallel reporter assays (MPRAs) (Peterman & Levine, 2016).

1.2 Sort-seq Background

Sort-seq involves sorting a population of input cells into tubes defined by gating on a phenotype measured at single cell resolution, most commonly fluorescence intensity level, followed by sequencing the DNA from those sorted cells. An overview of the Sort-seq workflow is shown in **Supplementary Figure 1**. This approach can be advantageous when a clinical phenotype or gene function may not be well correlated with cell proliferation and survival, and when fluorescent reporters are available to assess cellular function, such as reporters responsive to transcription factor activity

(Shepherdson et al., 2024). Beyond functional testing genetic variants in the context of human health, Sort-seq methods have also been used to monitor bacterial populations (Lambrecht et al., 2017) and to estimate variant impacts on protein properties such as stability and solubility (Gaber et al., 2024). Sort-seq generally requires the incorporation of a fluorescent reporter into the biological pathway being studied, which may require cell line engineering.

The Sort-seq experimental process produces a count distribution for each population of cells carrying any one of potentially thousands of variants. Cells are sorted into discrete bins, each spanning a select interval out of the full dynamic range of the perturbed fluorescence phenotype (shown as bins in **Supplementary Figure 1.4**). Resulting sequencing files are processed to map sequencing reads back to mutagenesis variants. The sequencing read counts from each sorted sample reflect the relative abundance of each variant. These processed read counts become inputs for downstream analyses, such as those performed in our Python package, *sortscore*.

1.3 Sort-seq Scoring Software

Compared to growth assays, analytical software support for scoring Sort-seq MAVEs remains comparatively limited but is improving. The Lilace R package began filling this gap by providing an implementation of a Bayesian hierarchical scoring framework to estimate latent unimodal shifts in the fluorescence distribution for variants relative to a negative control baseline (e.g. synonymous distribution) (Freudenberg et al., 2026).

We developed *sortscore*, which implements a weighted mean scoring approach to provide an additional, dedicated, reproducible tool for Sort-seq analysis. It incorporates experimentally derived bin weights—in the case of FACS, median fluorescence intensity (MFI)—when available, to retain quantitative detail and experimental units that are discarded when bins are treated as ordinal categories. Since MFI values are not strictly required, this approach can readily generalize to other bin-based MAVE assay designs beyond FACS-based Sort-seq. In its current implementation, *sortscore* processes sequencing read count data generated by FACS-based assays and provides activity scoring through a streamlined CLI workflow, along

with visualization options (64 codon heatmaps and tiled heatmaps) and tiled normalization for short-read sequencing contexts—features which are not currently implemented in other Sort-seq MAVE tools.

While sortscore and Lilace directly score individual variants via weighted mean and Bayesian hierarchical methods respectively, a branch of related tools models global genotype-phenotype landscapes from which variant-level phenotypes are then inferred. MAVE-NN is one such tool which accepts binned counts to train a neural network that deconvolutes measurement noise and nonlinearities (Tareen et al., 2022). LANTERN and MoCHI apply low-dimensional and mechanistic biophysical approaches, respectively (Tonner et al., 2022; Faure and Lehner, 2024). The methodological differences between these tools are described further in **Table 1** below.

Beyond these methodological distinctions, the customizable workflows implemented by sortscore lower barriers to Sort-seq analysis. Since weighted mean scoring has been commonly applied historically, sortscore facilitates reproduction of these prior analyses within a standardized framework. Its CLI implementation further provides an accessible option for researchers working in high-performance computing environments or incorporating scoring into a larger data pipeline.

2. Methods

2.1 Software Design

Standardized Analysis Workflows

Figure 1 (below) shows two standardized analysis workflows that can be executed using the command line interface. These console commands offer a convenient way to add scoring and visualization to the end of a sequencing data processing pipeline, as in compute environments with batch job schedulers. The Variant Scoring workflow takes processed sequencing counts files from Sort-seq experiments and returns a heatmap and a Score Set CSV file. Prior to normalization, batch effects can be apparent across Sort-seq experiments. The Batch Normalization workflow takes

Score Set files from multiple tiled experiments, and returns a combined Score Set and heatmap on a unified scale for comparison.

Dependencies

The sortscore package utilizes standard scientific Python libraries including NumPy and pandas for data processing, SciPy for statistical analysis, matplotlib and seaborn for visualization, and Biopython for biological sequence handling (Cock et al., 2009; Harris et al., 2020; Hunter, 2007; The pandas development team, 2026; Virtanen et al., 2020; Waskom, 2021).

Data Loading and Input Requirements

The sortscore package is focused on calculating variant scores based on relative abundance in sorted bins, normalizing data across tiled sorting experiments, and visualization of these results via heatmaps. Sortscore accepts variant counts files in TSV (gzipped or uncompressed) and Parquet formats, requiring that they contain two columns: sequence (or sequence identity) and unique counts. Currently available software is routinely used to convert raw sequencing files into the input variant counts file. Such tools include Enrich2 or DiMSum's WRAP module for short reads, or Pacybara for long reads (Faure et al., 2020; Rubin et al., 2016; Weile et al., 2024). Software for mapping barcodes to variants is available in the Call Variants DMS Docker container (*P-Ruzycki/Call-Variants-DMS-Docker*).

Variants with low read counts may be filtered during loading by specifying a threshold in units of reads per million in the configuration file. This minimum read filter applies at the bin level and is 0 by default.

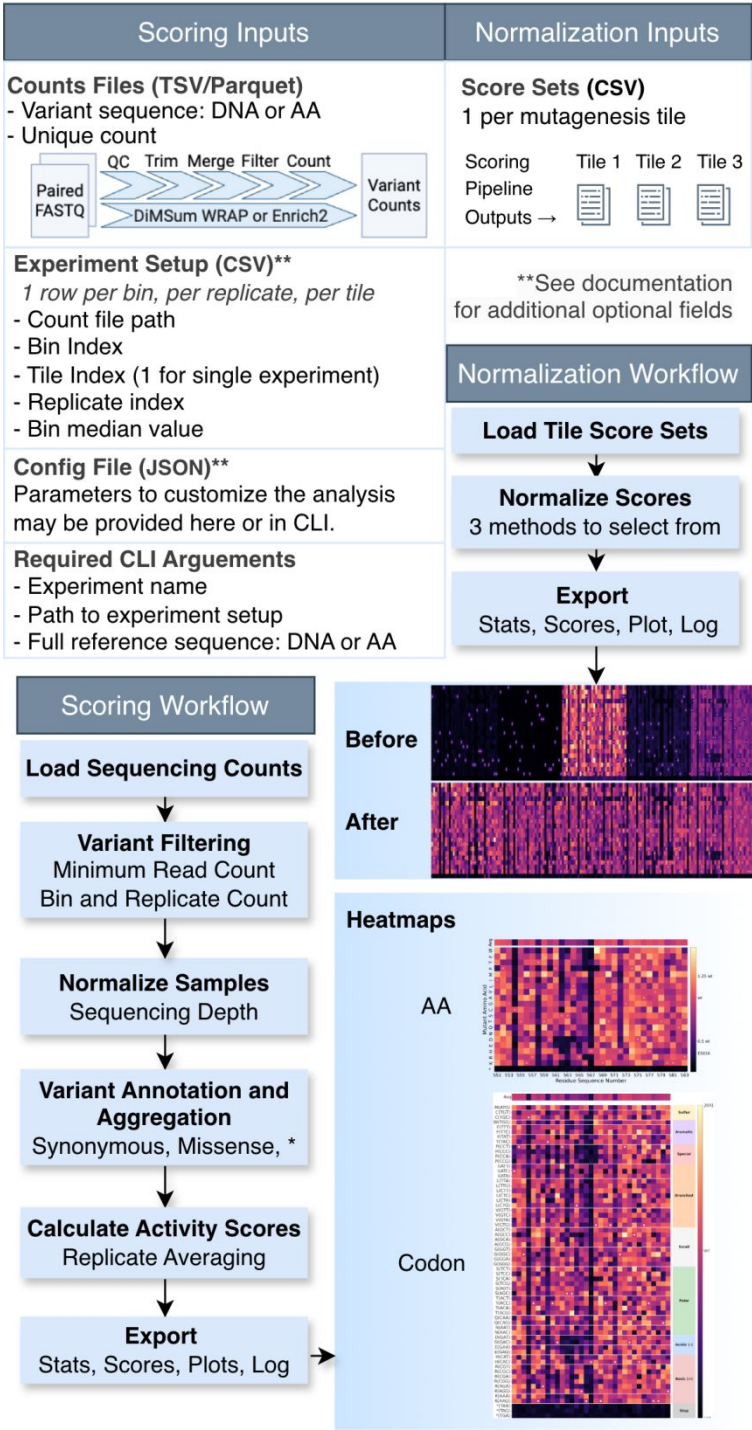


Figure 1: Workflow diagram for the sortscore package. Required inputs and key program steps are shown for both single experiment analysis for Variant Scoring (left) and batch analysis using Cross-Tile Normalization (right).

2.2 Analysis Configuration

Configuring Scoring for a Single Sorting Experiment

Easily modifiable configuration files provide flexibility to tailor the analysis workflows to new experimental assays using a basic text editor. Configuration for running sortscore is provided via two files: a) a JSON configuration file, and b) a CSV file to define the experimental design. Templates for each are available on GitHub: config.json and experimental_setup.csv, respectively. The project documentation on GitHub also lists the full set of required and optional fields available to customize the scoring workflow.

Modifiable fields in the JSON include the mutagenesis approach (e.g., all possible amino acids per position, presence or absence of nonsense variants), quality filter thresholds, and visualization customizations. The experiment setup CSV file provides the structure to share file paths, information about technical replicates, sum of read counts per sample, and gating strategy for a single sorting experiment.

Configuring Normalization Across Tiled Experiments

The same configuration file can be used to select the desired normalization method, specify negative controls to use during normalization, and provide paths to the configuration files for individual experiments that will be combined. The three available methods for normalization are described below. Prior to normalization across tiles, count files for each tile must be processed using the Variant Scoring workflow.

DNA and AA Variant Scoring and Heatmaps

Since a wide range of mutagenesis approaches are used in MAVE experimental designs, we provide a limited set of variant aggregation and heatmap customization options. When DNA sequences or variant identifiers are provided, the user can specify whether they want scores and heatmaps returned that reflect the DNA-level variant or protein sequences that have been aggregated across codons. The package auto-

detects the type of sequences present in the counts files and will alert the user if this is incompatible with the requested scoring option.

2.3 Variant Activity Scoring

The primary function of the first sortscore analysis pipeline is to return an activity score for each binned variant. To quantify activity, we estimate the mean phenotypic value utilizing elements from the discrete counts distribution for each variant. In Sort-seq, these distributions are produced by sorting cells into tubes corresponding to intervals of the measured range of phenotype values followed by sequencing. This simple estimator is practical, easy to understand and compute, and is useful for ranking variants and comparing resulting scores against the activity of benign-like and/or pathogenic-like control variants.

Normalization across Samples

The package controls for sequencing depth differences by normalizing total read counts across samples as reads per million (RPM). Total reads in a sample can be user-provided or can be calculated from the counts files.

$$\text{RPM}_{v,b} = \left(\frac{\text{read count}_{v,b}}{\text{total reads}_b} \right) \cdot 10^6$$

v = Variant index

b = Bin index

Score Calculation

For each variant, Sort-seq produces a discrete distribution defined by ordered fluorescence bins and the counts of variant sequencing reads observed in each bin, which can be modeled as a probability mass function. FACS discretizes the underlying continuous fluorescence distribution of the cell population containing a given variant (see **Supplementary Figure 2**). The sortscore package calculates an expected fluorescence score, S_v , as a weighted mean across bins, weighting the proportion of variant reads in each bin by a user-provided bin weight and summing across bins. This

weight is typically the bin MFI for experiments employing FACS, but any user-defined value can be substituted, including an integer-based ordinal value. MFI-based weights capture the actual, and often unevenly spaced, differences in intensity between bins, whereas ordinal weights assume evenly spaced ranks. Despite this simplification, we found ordinal-weighted scores to be highly concordant with MFI-weighted scores in a demo dataset with unevenly spaced bins (see **Supplementary Figure 3**). MFI-based bin weights produce an estimate of the variant-carrying cell population's mean fluorescence. Ordinal bin weights produce a comparable rank-based score directly from binned counts, without fluorescence units. Both options serve as proxies for variant function.

$$S_v = \sum p_{v,b} w_b$$

w_b = median fluorescence intensity or ordinal value assigned to bin b ,
 $p_{v,b}$ = proportion of reads for variant v in bin b after normalization for sequencing depth

Technical Replicate Averaging

By default, when averaging scores across technical replicates (defined as the same starting cell population sorted into bins in more than one FACS session), variant read counts serve as weights for calculating the mean across replicates. We use this weighting because we assume that higher read counts in a given replicate will result in more accurate scores for that replicate. The user can opt for a simple average instead in the analysis configuration.

$$\bar{S}_{v,\text{weighted}} = \frac{\sum_{r=1}^R S_{v,r} \cdot w_{v,r}}{\sum_{r=1}^R w_{v,r}}$$

$S_{v,r}$ = Activity score for variant v in replicate r

$w_{v,r}$ = Total normalized read count for variant v in replicate r

R = Number of replicates

Normalization and Scaling across Tiled Experimental Batches

Our package provides two options for reducing potential batch effects that may be observed across different tiled experiments, by which we mean the use of separate oligos that code for different portions of the same protein. These score adjustments to the activity scores bring them onto a similar scale for comparison.

Method 1 Linear range normalization

We provide this normalization method used in the 2018 publication of the BRCA1 saturation genome editing DMS (Findlay et al., 2018). An interval scale is set to the score difference between global nonsense (zero) and global synonymous score (one) to bring batches into alignment and aid interpretability.

$$S_{\text{norm}} = b \left(\frac{A - C}{a - c} \right)$$

b = Variant activity score

$\left(\frac{A - C}{a - c} \right)$ = Scaling factor (global range / local range)

a = Synonymous median (local)

c = Nonsense median (local)

A = Synonymous median (global)

C = Nonsense median (global)

Method 2: Biological two-control anchoring, “two-pole” (Default)

This normalization method (see **Supplementary Figure 4**) is the default used by sortscore and assumes that the tile-specific synonymous variant scores and nonsense variant scores should be similar to one another across experiments. It involves a three-step process: 1) normalization by the ratio of the global median (M_{syn}^{global}) and the tile-specific median (M_{syn}^{tile}) of synonymous variants, which all code for the same wildtype protein, 2) standardization of scores based on the scale of the distribution of synonymous scores, and 3) normalization by the ratio of the global and tile-specific medians of nonsense variants. The units of scores resulting from this method do not represent either the relative fold change in activity or the unit variance from a particular

distribution, but rather should be interpreted in reference to positive and negative controls, when possible. However, this method does cause the directionality of activity change to be immediately apparent, because the mean synonymous score is centered at zero.

$$S_v^{(1)} = S_v \cdot \frac{M_{syn}^{global}}{M_{syn}^{tile}}$$

$$S_v^{(2)} = \frac{S_v^{(1)} - \bar{S}_{syn}^{(1)}}{\sigma_{S_{syn}^{(1)}}}$$

$$S_{norm,v} = S_v^{(2)} \cdot \frac{M_{nonsense}^{global}}{M_{nonsense}^{tile}}$$

Method 3: Biological single control anchoring, “one-pole”

This method is equivalent to Method 2, except that it omits the nonsense normalization factor to accommodate experiments that may not include multiple nonsense variants in each experimental batch. Instead, only the synonymous scaling factor is applied to scores by multiplying by the ratio of the global synonymous median and the tile-specific synonymous median.

Quality Control Filtering

In an effort to improve the quality of the resulting scores, we provide options for filtering out or dropping values that may not meet certain thresholds. For example, a minimum replicates filter allows users to specify the number of technical replicates that each variant should be present in so that it can be included in the final score set of values averaged across replicates. The rationale for this filter is that if a variant is missing from multiple technical replicates, it may be poorly represented in the studied population, and any resulting scores may be unreliable due to insufficient sampling of that variant. A related minimum bins filter requires that a variant is represented in a given number of sorted samples in order to be assigned a replicate level score. This filter is highly context dependent because some experimental designs will intentionally

yield zero counts in some bins, but as a guide one might consider the number of bins that control variants fall into. The default behavior is to not apply these filters.

2.4 Data and Visualization Outputs

Plotting Capabilities

A potentially particularly valuable feature to users is that sortscore produces codon and protein substitution heatmaps to visualize relative activity scores across variants. Position within the protein is oriented on the y axis and each variant is represented by a different row on the x axis. Optional biophysical property annotations may aid in interpretation of calculated scores. Positional average scores and coverage statistics are also included in the heatmap visualizations. Plotting functions can save outputs in PNG, PDF, or SVG formats.

Data Exports

Sortscore also returns full Score Sets, Score Set summary statistics, and the matrix of scores that are displayed in the heatmap. The heatmap matrix values are provided so the user can easily recreate and reformat the plot using other tools. This output will be raw scores if sortscore was run separately for individual experiments, or adjusted scores if run in batch mode. Summary statistics are also provided for different classes of variants in a JSON file, such as synonymous with wildtype versus nonsense versus all missense variants, allowing rapid assessment for how the scores were distributed in a given experiment.

2.5 MAVE Ecosystem Integration

MaveDB serves as a centralized repository for functional datasets and provides visualization functionality, with upcoming ClinVar submission integration (Esposito et al., 2019). Structured JSON configuration files and detailed output and error logging in sortscore enhance its interoperability with web applications, enabling potential integration with community resources such as MaveDB.

3. Results

3.1 Processing Sort-seq Data from GLI2 DMS

The data shown in the heatmap in Figure 1 was produced in a series of DMS experiments performed by our lab with the aim of characterizing the pathogenicity of variants within the GLI2 transcription factor DNA binding domain, from a manuscript in preparation. Additional experimental details are provided in **Supplementary Figures 1 and 2**.

A fluorescent reporter integrated into the genome of the NIH3T3 cell line produced GFP as a readout of intact transcriptional activity (Pusapati et al., 2018). GFP (FITC) fluorescence intensity was used to divide the cell population into four equal bins, each representing approximately 25% of the GFP expression distribution (see **Supplementary Figure 2**). DNA was then extracted and sequenced, and flow cytometry data were analyzed using FlowJo software.

Counts data were produced from FASTQ files using standard bioinformatics tools: cutadapt for trimming adapters (Martin, 2011), VSEARCH for merging paired reads (Rognes et al., 2016), and a custom awk script used to confirm the read was a member of the list of designed sequences. As expected, the sortscore analysis prominently displays the deleterious impact of variants in the zinc finger domain as vertical dark stripes on the heatmap, particularly for the residues known to chelate the zinc ions (Pavletich & Pabo, 1991).

The use of a sequential colormap for the heatmap improved interpretability of these GLI2 data compared to the standard red-blue versions commonly seen in MAVE publications which have great utility in the presence of both gain and loss of function missense variants. However, we have found no definitive evidence for gain of function behavior in the GLI2 DNA binding domain DMS to date. When only one prominent direction of effect is present, as in these data, sequential colormaps can make critical residues more apparent. The complete dataset will be made available upon publication of a manuscript currently in preparation, which also utilized sortscore for analysis.

We also tested the package by scoring an existing DMS dataset for the CRX gene using default options (Shepherdson et al., 2024). The resulting heatmap is shown in **Supplementary Figure 5**.

3.2 Tool Comparison

Table 1 considers publicly available software packages that address the need for variant scoring in Sort-seq experiments. These separate broadly into two approaches, 1) genotype-phenotype landscape modeling methods that first model a surface for the region then predict latent phenotypes as a function of this feature space and 2) methods directly utilizing individual variant fluorescence distributions from experimental binned counts to produce estimates. The sortscore package adds to this ecosystem by providing an intuitive and direct activity score calculation in a user-friendly package. When the experimental bin median values are known, scores are returned in the original scale of the experiment that generated the data. Of the tools considered, Lilace is the closest comparator to sortscore in its goal of providing a dedicated tool for scoring FACS-based DMS data; we therefore compared sortscore estimates to those from Lilace on a FACS-based deep mutational scan of GLI2 residues 518–550, finding strong agreement between the two methods (Spearman's $\rho = 0.99$, $n = 693$; **Supplementary Figure 6**). In **Supplementary Table 1**, we summarize additional tools, primarily designed for growth assays, that are presently available for MAVE analysis and may be of interest to the reader.

Table 1: Comparison of selected MAVE analytical tools that provide variant-level activity scoring or phenotype prediction for Sort-seq experiments.

Software	Type	Summary	Variant Score / Phenotype Prediction	Uncertainty Quantification
sortscore	Variant Scoring from binned counts	Quickly score variants and align tiled Sort-seq experiments	MFI bin weights produce a variant fluorescence estimate; ordinal bin weights produce a rank-based score	Not presently implemented
Lilace		Robust FDR control for DMS employing FACS	Latent unimodal shift in fluorescence distribution relative to negative control	Negative control-based bias correction; local false sign rate (lfsr)
MAVE-NN	Genotype-Phenotype Landscape Modeling	Learn quantitative G-P maps and deconvolute noise and nonlinearities	Output of the nonlinear function transforming the latent phenotype into measurement units	Parametric bootstrap approach
LANTERN		Find low-dimensional feature space for interpretable discovery	Posterior Mean of the nonlinear surface	Posterior predictive uncertainty
MoCHI		Fit mechanistic models to quantify absolute energies and interactions	Predicted observed phenotype that converts molecular phenotype into units of assay	Monte Carlo confidence intervals

4. Limitations and Future Development

4.1 Opportunities for Improvement

Sortscore is focused on addressing scoring and visualization needs for Sort-seq experiments and avoids redundant implementation of functionality already provided by other, more established MAVE software. At present, users must generate input counts from raw sequencing data prior to running sortscore, though future integration work could streamline handoff from tools that perform this upstream step. This initial release

implements an easy-to-interpret weighted average and basic quality filtering. Because this initial implementation computes scores directly rather than fitting a statistical model, uncertainty quantification is not built into the estimation method; as a result, sortscore outputs are best used as inputs for downstream MAVE interpretation rather than as a final analysis endpoint at this time. Additional next steps may include identifying variants with activity distinguishable from a reference background distribution, comparing score distributions across functional classes, integrating external annotations, and prioritizing variants for follow-up testing. Toward this end, sortscore already calculates a modified z-score during batch normalization, and this internal machinery could be extended in future work to provide a formal per-variant significance test, such as a p-value distinguishing variant activity from background.

Examples of analysis approaches that are adjacent to the sortscore package include the Oddspath calculation which can be used to determine an overall assessment of a functional assay’s ability to accurately discriminate benign from pathogenic variants (Brnich et al, 2018), and ExCALIBR, which can assign evidence levels to individual functional scores (Tejura et al., 2026). At a higher level, community-maintainable and reproducible approaches that bundle MAVE analysis tools might include containerizing complementary software via Docker or developing Nextflow modules that could be used with the deepmutscan nf-core pipeline (Wehnert & Stammnitz, 2025).

4.2 Looking Beyond FACS: Applicability across Cell Sorting

Unexplored and underutilized sortable cellular phenotypes show potential for adding orthogonal layers of evidence to atlases of variant effects. Additional multiplexed measurement modalities compatible with sequencing would be valuable for directly quantifying variant effects and mapping their impact onto underlying biological mechanisms. Current sorting approaches coarsen phenotypic data, introducing significant noise at this stage of Sort-seq experiments. Throughput of millions of cells containing thousands of variants is possible in a single experiment, but even so, profiling variants using Sort-seq techniques remains costly, both in time and reagent /

equipment expense. Yet the clear conceptual throughline from MAVE data generation to diagnostic yield primes these techniques to be substantially sharpened in the coming years.

Sortscore could be applied to score any continuous, sortable phenotype, so long as the technique doesn't degrade the binned genetic material. This package could provide an easily accessible scoring framework for up and coming technologies that may expand what we refer to as Sort-seq. Sortable and multiplexable phenotypes might include additional spectroscopic, electrical, mechanical, or morphological cellular features. Responsibility lies with the user to determine if such a metric adequately captures the genotype-phenotype relationship.

4.3 Concluding Remarks

We provide a clearly documented and accessible tool to serve as a starting point for analyzing Sort-seq counts data. Questions, bug reports, and feature requests may be directed to the project's GitHub Issues page at <https://github.com/dbaldrige-lab/sortscore>. By releasing this package, we hope to lower the barrier to entry for groups exploring these experimental methods. We also hope to make validating the conclusions gleaned from Sort-seq equally approachable to groups that do not routinely use these methods, which may increase parity in the analysis protocols for Sort-seq variant effect assays compared to the more widely employed growth-based methods in use today.

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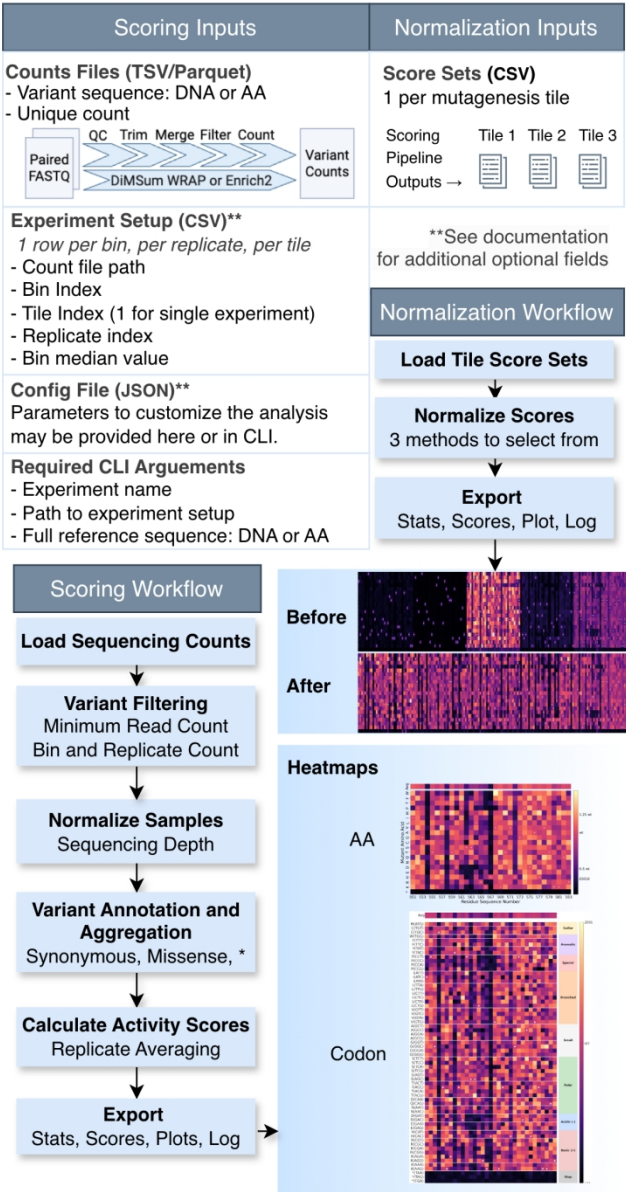
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